

Fig. 5: Pin details of DB25 connector (Courtesy: www.omega.com)

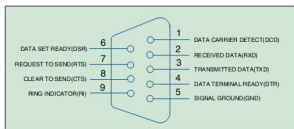


Fig. 6: Pin details of DB9 connector

corresponding to the clock frequency of 50MHz (50,000,000). The Parity and the Even constants are set as True and False, respectively. The constant Baud is assigned a value 19200 corresponding to baud rate of 19.2 kbps. The two constants MaxFactor and Divisor2 have values 2604 (Fxtal/Baud) and 1302 (Fxtal/Baud/2), respectively. All the constants mentioned above will be used in the transmitter FSM as well as in the receiver FSM.

Next in the VHDL code have been defined the signals used in the transmitter FSM followed by the signals used in the receiver FSM. They have been defined separately for ease of the readers. Signal TxFSM is of type TxFSM_State having five states, namely, Idle, Load_Tx, Shift_Tx, Parity_Tx and Stop_Tx. The initial state of the signal TxFSM is Idle. Other signals defined are Tx_Reg which is a 9-bit std_logic_vector signal used as a transmit register. TxDivisor and TxDiv are

integers that can have values ranging from 0 to 2604 and TxBitCnt is also an integer having range from 0 to 15. TopTx and Tx_Par are std_logic signals.

The signals used in the receiver FSM are RxFSM, Rx_Reg, RxDivisor, Rx_Par and Rx_r. RxFSM is a signal of type RxFSM_State having eight states, namely, Idle, Start_Rx, Shift_Rx, Edge_Rx, Parity_Rx, Parity_2, Stop_Rx and RxOVF. RxFSM is in the initial state, that is, the Idle state. Rx_Reg is an 8-bit receive register used to store the received serial data. Signals RxDivisor and RxDiv are integers having values ranging from 0 to 1302. RxBitCnt is an integer having values ranging from 0 to 15. TopRx, RxRDYi, Rx_Par and Rx_r are std_logic signals.

The first line after the 'begin' statement assigns the value of RxRDYi to RxRDY. RxRDY is the flag indicating that data is available during reception. The first process is used for receiver input resynchronisation. It has RST and CLK in the sensitivity list. If RST is at logic level '1', the resync signal Rx_r is set to logic level '1', else on the rising edge of the input clock (CLK) the value of the serial data in (Rx) is sent to Rx_r.

The next process is used for baud rate conversion. This process also has RST and CLK in the sensitivity list. When RST is at logic level '1', the values of RxDivisor and TxDivisor are set to '0', else on the rising edge of the clock, RxDivisor and TxDivisor are assigned values $[(Divisor2 - 1) = 1301]$ and $[(2 \times Divisor2) - 1] = 2603$, respectively.

The next process is used for receiver clock generation. The receiver clock has a time period of bit time/2. The process has the RST and CLK signals in the sensitivity list. When RST is at logic level '1', the values of RxDiv and TopRx are set to '0'.

RxDiv is the receive clock generation register and TopRx is the receive master clock pulse. If the RST is at logic level '0', on the rising edge of the clock CLK, TopRx is set to '0'. If the RxFSM signal is in the initial Idle state, RxDiv is also set to '0', otherwise if RxDiv is equal to RxDivisor (1301), it is set to '0' and TopRx is set to '1', else value of RxDiv is incremented by one. The process generates a pulse having an on time of $1/50 \times 10^9 \text{ s} = 20 \text{ ns}$ and a frequency of 38.4kHz whenever the RxFSM signal is not in the Idle state.

The next process generates the transmitter clock having the same time period as the bit time. The process is similar to the one used for generating the receiver clock pulse. It generates a clock pulse having 20ns width at a frequency of 19.2kHz. The major difference between the transmitter and the receiver clocks is that the transmitter clock runs continuously and does not depend upon the state of the FSM.

The transmitter FSM is implemented next. The LSB of the transmit register Tx_Reg is transferred as serial data out to the Tx pin. The transmitter FSM is implemented in the Tx_FSM labelled process. It has RST and CLK in the sensitivity list. When RST pin is at logic level '1', the transmit register is set to '1FF', the state of TxFSM signal is set to Idle and TxBitCnt, TxBusy and Tx_Par are all set to '0'. If the RST pin is at logic level '0', on the rising edge of the clock CLK, TxBusy is set to logic